Vectors, lines and planes in space

Lecture and worked examples

Objectives

Calculate with vectors; use collinearity and coplanarity; characterise lines and planes; work in a frame.

Vector calculus

Chasles' relation $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$, the sum of vectors and multiplication by a real number extend to space. Two non-zero vectors are collinear if one is a multiple of the other. The zero vector is collinear with every vector. Three vectors are coplanar if they belong to the same vector plane. If $\vec{u}, \vec{v}$ are not collinear, then $\vec{u}, \vec{v}, \vec{w}$ are coplanar if and only if $\vec{w} = \alpha\vec{u} + \beta\vec{v}$ for some real numbers α, β.

The line through A with non-zero direction vector $\vec{u}$ is the set of points M such that $\overrightarrow{AM} = t\vec{u}$ for some real number t. The plane through A spanned by two non-collinear vectors $\vec{u}, \vec{v}$ is the set of points M such that $\overrightarrow{AM} = s\vec{u} + t\vec{v}$ for some real numbers s, t.

Three non-collinear points define a plane. Four points A, B, C, D, with A, B, C not collinear, are coplanar if and only if $\overrightarrow{AD}$ is a linear combination of $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

Relative positions

Two lines are coincident, strictly parallel, intersecting or non-coplanar (skew). Collinear direction vectors characterise lines that are parallel in the broad sense. Non-collinear direction vectors do not guarantee an intersection. A line and a plane: line contained in the plane, parallel and disjoint, or meeting at a single point. Two planes: coincident, strictly parallel, or intersecting along a line.

Relative positions

In $ABCDEFGH$, the faces $ABCD$ and $EFGH$ are parallel and the edges AE, BF, CG, DH correspond. The lines (AB) and (CG) are neither parallel nor intersecting: they are non-coplanar.

Basis and coordinates

A basis of space is a family of three non-coplanar vectors. Every vector can be written in it in a unique way as $x\vec{\imath} + y\vec{\jmath} + z\vec{k}$. A frame consists of an origin O and a basis.

$$\overrightarrow{AB} = (x_B - x_A, y_B - y_A, z_B - z_A), \qquad I = \left(\frac{x_A + x_B}{2}, \frac{y_A + y_B}{2}, \frac{z_A + z_B}{2}\right).$$

In an orthonormal frame, $\|\vec{u}\| = \sqrt{x^2 + y^2 + z^2}$ and
$AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$.

Basis and coordinates

In an orthonormal frame, for $A(1,0,2)$, $B(3,-1,4)$, $\overrightarrow{AB} = (2,-1,2)$, $AB = 3$ and the midpoint is $(2,-\frac{1}{2},3)$.

Basis and coordinates

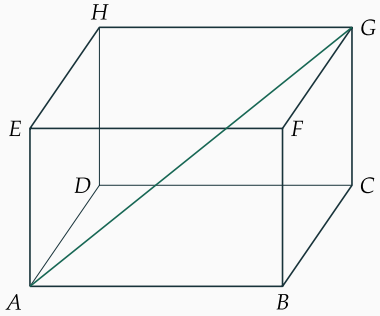
Write the three coordinate equalities. Determine the coefficients from two equations when possible, then check the third. A contradiction proves non-membership; do not discard the inconvenient equation.

Research activity

Three midpoints in a cube - 40 min.

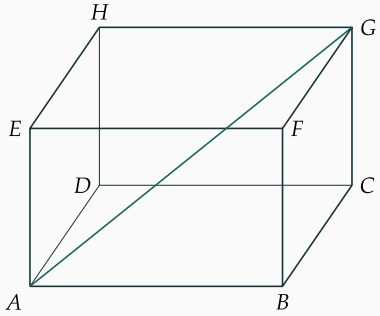
Consider the cube $ABCDEFGH$ shown below. Let $\vec{i} = \overrightarrow{AB}$, $\vec{j} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points I, J and K are the midpoints of $[AB]$, $[AD]$ and $[AE]$ respectively. Let O be the centre of the cube. $[AG]$ is a space diagonal, drawn as a green dashed line.

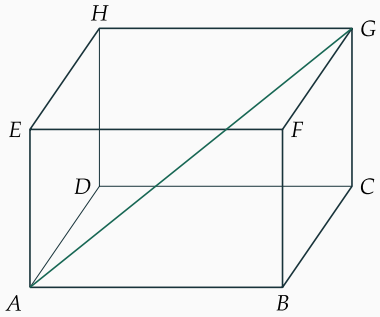
The hidden edges are dashed blue; D and F do not lie on (AG).



Research activity (2)

1. Express $\overrightarrow{IJ}$ and $\overrightarrow{IK}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$.
2. Prove that the points I, J and K are not collinear.
3. Determine whether O belongs to the plane (IJK). Justify the answer by investigating whether $\overrightarrow{IO}$ is a linear combination of $\overrightarrow{IJ}$ and $\overrightarrow{IK}$.



Research activity (3)

4. Why does simply reading the figure not allow one to prove that O does or does not belong to the plane (IJK)?